A Distribution and Structure Match Generative Adversarial Network for SAR Image Classification

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***Abstract*— Synthetic aperture radar (SAR) image classification is a fundamental research in the interpretation of SAR images. The previous methods are unilaterally based on statistical features or spatial features, which cannot capture features with complete SAR image characteristics and unavoidably limits the performance for classification. In this article, novel sample weighting and class adversarial training strategies are proposed to fuse complementary SAR characteristics. Based on these, a distribution and structure match auxiliary classifier generative adversarial network (DSM-ACGAN) is constructed for high-quality discriminative feature learning. Particularly, the characteristics of statistical distribution and spatial structure are jointly considered in class adversarial training of DSM-ACGAN. On the one hand, DSM-ACGAN sets the true SAR image characteristics as goals for the generator to learn generative models of each category. On the other hand, and more importantly, it guides the discriminator to simultaneously capture the desired statistical and structural features. Through the class adversarial processing, the discriminative feature learning progressively improves and contributes to classification. Additionally, class-balanced and plausible samples can be generated. Experimental results on three broad SAR images from different satellites confirm the effectiveness of class adversarial training and the superiority of discriminative feature learning in DSM-ACGAN. Visual performance and quantitative metrics also show the state-of-the-art performance of the novel model.**

***Index Terms*— Discriminative feature learning, generative adversarial network (GAN), image classification, synthetic aperture radar (SAR).**

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I. Introduction

As an active microwave imaging sensor, synthetic aperture radar (SAR) [1] has the ability to observe the ground condition without the constraints of illumination and the atmosphere. Therefore, it is widely used in civil and military applications. SAR image classification is an essential step in SAR understanding and interpretation and has a significant effect on the study of change detection [2], [3], urban detection [4], environment and resources monitoring [5], etc. In particular with the rapid increase of data, accurately classifying high-resolution SAR images becomes an urgent requirement.

Feature extraction or feature learning plays an important role in SAR image classification. Sometimes, it can almost be said that the discrimination of learned feature decides the performance of classification. Thus, plenty of work has been done on how to effectively extract features. From the view of extracted features, there are two main classifications, i.e., scattering-based statistic modeling and content-based spatial feature extraction and learning.

Due to its unique scattering mechanism, the distribution of pixel values within SAR images provides much valuable information. An accessible way to characterize SAR images is statistically modeling the terrain distributions. First, several single parametric probability density models like lognormal, Weibull, negative exponential distribution [6], Fisher [7], generalized Gamma [8], Rayleigh [6], Nakagami-Gamma [9], heavy-tailed Rayleigh [10], generalized Gaussian Rayleigh [11], $\mathcal{K}$ [12], and $\mathcal{G}$ [13] are proposed to describe the distribution characteristics of SAR images in specific cases. To fully capture the complex content of higher resolution SAR images, more extensible methods arise one after another, such as Gamma mixture model [14], Rayleigh mixture model [10], dictionary-based mixture density estimation [15], [16], beta generalized normal distribution [17], Gaussian mixture model [18], mixture of Alpha-stable model [19], Cauchy–Rayleigh mixture model [20], Student's t-mixture model [21], K-lognormal mixture model [22], and generalized gamma mixture model [23]. These models are then used in a Bayesian inference framework to realize classification. But they are constructed more or less under strict assumptions or for particular scenes, which greatly limits the accuracy and the field of application. In addition, these models are always based on pixel-level features, which lack the ability to capture the high-level semantic relationship.

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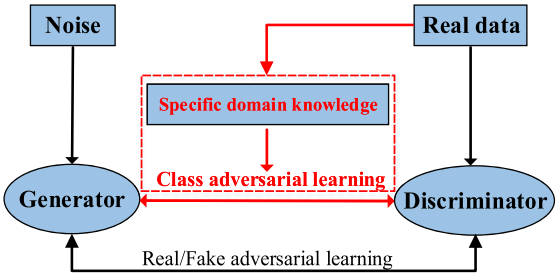


Fig. 1. Framework of dual adversarial learning. The proposed novel processing method are in red box.

Content-based spatial feature extraction and learning is another important means to describe SAR images. In the beginning, the texture-based methods such as gray-level co-occurrence matrix [24] hold dominant position to obtain the pattern regularity of different terrains. Motivated by the development of analytical geometry, transformation domain methods, for instance, Gabor transform [25], wavelet transform [26], curvelet transform [27], [28], and contourlet transform [29] have been constructed to characterize spatial locality, scale, and orientation of pixels. With the increasing of spatial resolution in SAR images and the development of algorithms in machine learning, massive mid-level and high-level features are utilized to describe the content of SAR images. Bag-of-words [30], [31], sparse representation [32]–[34] and hierarchical representation [35]–[37] adequately excavate low-level information and integrate them to give a description of the image content in a thematic level. Recently, the methods of deep learning grow vigorously in remote image processing. The works [26], [38]–[47] adopt deep neural networks to extract high-level spatial features for classification. The works [48]–[53] have confirmed the potential of generative adversarial network (GAN) in SAR image processing, and they mainly focus on generating data with the generator. In the feature learning process of the above content-based methods, the statistical properties of SAR images are rarely taken into account, which leads to the ambiguity of learned features and reduces their discrimination for classification.

From the above description, the low-level statistical modeling and no guidance of high-level feature learning bring about the insufficient discrimination of learned features. To promote comprehensive and in-depth discriminative feature learning, a dual adversarial learning framework (shown in Fig. 1) is proposed, which consists of the traditional real–fake adversarial learning and the proposed class adversarial learning. Accordingly, to achieve this for SAR image classification, a model called distribution and structure match auxiliary classifier GAN (DSM-ACGAN) (shown in Fig. 2) is constructed.

According to the statement that specific domain knowledge can be used to help design representations [54], a dual adversarial learning framework is assumed. It can impose class-specific domain characteristics to the adversarial training for more effective confrontation and improve the discriminative feature learning. Specifically, during class adversarial training, the distribution and structure properties of real data are utilized to evaluate and boost the generative and discriminative ability of the model. If the generated samples are not plausible or the discriminator cannot map the input to a class-separable feature space, the loss is modified through the sample weighting scheme and the training continues. Finally, the generator can automatically capture accurate models of real data and generate plausible data without artificial design. Meanwhile, the discriminator is able to obtain strong discriminative features through the confrontation with captured statistical and structure characteristics from the generator. Accordingly, DSM-ACGAN achieves automatic and accurate distribution modeling and explicit feature learning with SAR statistical characteristics. The pixel-level statistical features and semantic-level spatial features are integrated to enhance the discrimination of learned features and improve the classification performance.

The main contributions of this article are as follows.

1) A dual adversarial learning framework is presented to fuse class adversarial learning to real–fake adversarial process of GAN, thus producing more effective confrontation and enhancing the ability of representation learning.
2) A novel network called DSM-ACGAN is constructed for a high-resolution SAR image classification, which exploits real–fake and class adversarial cotraining guided by complementary SAR characteristics to learn high-level discriminative features from raw SAR images.
3) A sample weighting scheme is developed to promote class adversarial training in DSM-ACGAN. It also guides the modeling in the generator from the aspects of statistical distribution and spatial structure, generating high-quality and class-balanced SAR images.

The remainder of this article is arranged as follows. Section II gives some propaedeutic. Section III is an elaborate description of our proposed method. Experimental results and analysis are shown in Section IV. Finally, the conclusion and future work are drawn in Section V.

II. Propaedeutic

This section provides propaedeutic, including the metric of statistical distribution, the metric of spatial structure and ACGAN [55].

A. Metrics of Statistical Distribution

According to [23], two metrics are chosen to assess the ability of statistical modeling of DSM-ACGAN.

The first one is Kolmogorov–Smirnov distance [56], which represents the maximum absolute difference between the fit cumulative distribution function $F(t_n)$ and the empirical cumulative distribution function $\hat{F}(t_n)$ as follows:

$$D_{\mathrm{KS}} = \max_{t_n \in \Omega} |F(t_n) - \hat{F}(t_n)| \tag{1}$$

where t refers to the gray level. t_n is the nth gray level. Ω is the set of gray level. The number of gray level determines the precision of the distribution curve. In our experiments, the gray level is 256.

The second one is symmetric Kullback–Leibler divergence [57], which measures the dissimilarity between the

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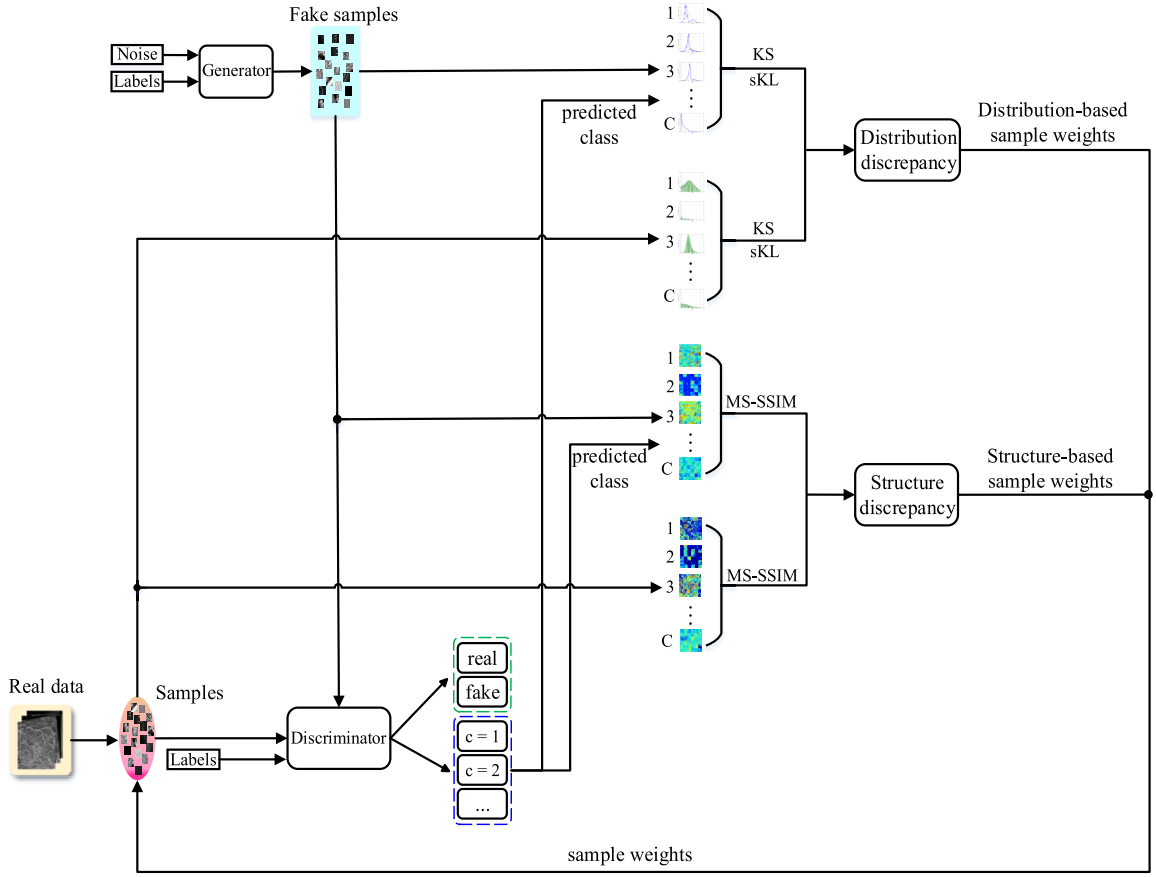


Fig. 2. Flowchart of DSM-ACGAN. During each epoch of training, the noise in latent space is first projected to data space, making the fake samples generated. Then fake samples and real samples are fed to the discriminator for discriminative feature learning and class prediction. Next, the class-specific distribution and structure discrepancy between the real and fake samples are assessed. Finally, two kinds of sample weights are combined to compose the class weights, which are imposed on the real samples in the next epoch. C represents the number of terrains.

estimated probability distribution function $f(t_n)$ and the normalized histogram $g(t_n)$ as follows:

$$D_{\text{skl}} = \sum_{t_n \in \Omega} f(t_n) \log \frac{f(t_n)}{g(t_n)} + \sum_{t_n \in \Omega} g(t_n) \log \frac{g(t_n)}{f(t_n)}. \quad (2)$$

In both metrics, a small value indicates a better fit of the particular distribution to empirical data.

B. Metric of Spatial Structure

Structural similarity (SSIM) [58] is a perception-based model, which is for image quality evaluation. It is not only based on the degradation of structural information but also incorporates important perceptual phenomena like luminance and contrast masking. Let $\mathbf{x}$ and $\mathbf{y}$ represent two nonnegative signals, respectively, then luminance, contrast and structure comparison measures are given as follows:

$$l(\mathbf{x}, \mathbf{y}) = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1} \quad (3)$$

$$c(\mathbf{x}, \mathbf{y}) = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2} \quad (4)$$

$$s(\mathbf{x}, \mathbf{y}) = \frac{\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3} \quad (5)$$

where μ_x and μ_y are the means of $\mathbf{x}$ and $\mathbf{y}$ respectively, σ_x^2 and σ_y^2 are the variances of $\mathbf{x}$ and $\mathbf{y}$ respectively, σ_{xy} is the covariance of $\mathbf{x}$ and $\mathbf{y}$, measuring the tendency of $\mathbf{x}$ and $\mathbf{y}$ to vary together. $C_1 = (k_1 L)^2$, $C_2 = (k_2 L)^2$ and $C_3 = C_2/2$ are the three variables to stabilize the division with a weak denominator. L is the dynamic range of the pixel-values (typically this is $2^{\text{bits/pixel}} - 1$, e.g., $L = 255$ for 8 bits/pixel gray scale images). $k_1 = 0.01$ and $k_2 = 0.03$ by default. The general definition of SSIM is

$$\text{SSIM}(\mathbf{x}, \mathbf{y}) = [l(\mathbf{x}, \mathbf{y})]^{\alpha} \cdot [c(\mathbf{x}, \mathbf{y})]^{\beta} \cdot [s(\mathbf{x}, \mathbf{y})]^{\gamma} \quad (6)$$

where α, β, and γ are parameters to define the relative importance of the three components.

To provide more flexibility, multiscale SSIM (MS-SSIM) [59] is proposed to incorporate the variations of image resolution and viewing conditions. It iteratively filters and downsamples the input image and finally obtains multilevel structure information. This procedure can be expressed as the following formula:

$$\text{MS-SSIM}(\mathbf{x}, \mathbf{y}) = [l_M(\mathbf{x}, \mathbf{y})]^{\alpha_M} \cdot \prod_{j=1}^{M} [c_j(\mathbf{x}, \mathbf{y})]^{\beta_j} \cdot [s_j(\mathbf{x}, \mathbf{y})]^{\gamma_j} \quad (7)$$

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where M is the scale, and luminance comparison is computed only at the Mth scale and denoted as $l_M(\mathbf{x}, \mathbf{y})$. At the jth scale, contrast comparison and structure comparison are calculated, which are denoted as $c_j(\mathbf{x}, \mathbf{y})$ and $s_j(\mathbf{x}, \mathbf{y})$ respectively. α_M, β_j, and γ_j are three parameters to define the relative importance of different scales, whose values are shown in Section IV-A.

C. ACGAN

Considering the ideal structure to combine generative modeling and discriminative feature learning, GAN [60] is chosen as the basic model, which consists of two parts: the generator and the discriminator. The generator learns to map a random vector $\boldsymbol{z}$ *(in the latent space* p_z*)* to target space, whereas the discriminator takes both fake and real data as input and outputs higher confidence if it distinguishes a sample is more likely to be real. This adversarial learning process can be regarded as a game between counterfeiter and police officer until Nash equilibrium is reached. Assuming the distribution of pixel values within real and generated images is p_r and p_g, respectively. G and D represent the generator and the discriminator, respectively. $\boldsymbol{x}$ is the real sample, and $V(D, G)$ is the value function. Then the game can be formulated as the following min–max problem:

$$\min_G \max_D V(D, G) = E_{\mathbf{x}\sim p_{r(\mathbf{x})}}[\log D(\mathbf{x})] + E_{\mathbf{z}\sim p_z(\mathbf{z})}[\log(1 - D(G(\mathbf{z})))]. \quad (8)$$

The baseline architecture of GAN in this article is ACGAN. Different from GAN, ACGAN leverages additional class side information and auxiliary class decoder to further fit the input. Specifically, the class labels are added to both the generator and the discriminator to produce class conditional samples and reconstruct side information. Correspondingly, the discriminator simultaneously distinguishes the resource and class of the input. The objective function consists of the following two parts: the log-likelihood of the correct source L_S and the log-likelihood of the correct class L_C

$$L_S = E[\log D(\mathbf{x})] + E[\log D(G(\mathbf{z}))] \quad (9)$$

$$L_C = E[\log D(C = c|D(\mathbf{x}))] + E[\log D(C = c|D(G(\mathbf{z})))]. \quad (10)$$

The generator is trained to maximize $L_C - L_S$, whereas the discriminator is trained to maximize $L_C + L_S$.

III. Method

In this section, a novel sample weighting scheme is adapted to ACGAN to develop a generative and discriminative framework named DSM-ACGAN. The pipeline of the proposed approach is shown in Fig. 2. A fundamental description of the sample weighting scheme is represented in Section III-A, and Section III-B is an analysis of the corresponding class adversarial training. Section III-C gives a specific presentation of discriminative feature learning in DSM-ACGAN.

In the test phase, the discriminator and its auxiliary classifier are used to predict the class of unseen SAR images.

A. Sample Weighting Scheme

Adjusting sample weights is a simple and effective method to emphasize the importance of certain samples and affect the training. In this article, the discrepancy derived from statistical distribution and spatial structure is used as sample weights, which reflect the quality of generative modeling in the generator. More importantly, it is used to guide better modeling of real samples during class adversarial training. A particular description of the scheme is given below.

Distribution characteristic is one salient feature in SAR images, representing the components in a complex scene. From this aspect, a distribution-based sample weighting term is constructed to constrain the distribution of pixel values within the generated images. Let W^c_{ks} and W^c_{skl} represent KS distance and sKL divergence between the generated images and real ones in cth category, respectively, where $c \in \{1, 2, \ldots, C\}$. Then the total distribution discrepancy W^c_d of category c is calculated as follows:

$$W^c_d = 0.5 \cdot \left(W^c_{\text{ks}} + W^c_{\text{skl}}\right). \quad (11)$$

As described in Section II-A, Kolmogorov–Smirnov distance characterizes the discrepancy from a global view, whereas symmetric Kullback–Leibler characterizes the divergence from a local view. Formula (11) integrates both and becomes a comprehensive metric of statistical modeling.

Due to the disparate imaging mechanism from natural images, the high-resolution SAR image has its special texture structure and spatial arrangement, such as foreshortening, shadow, and layover. The rich spatial information should be contained in generative modeling. Hence, a structure-based sample weighting term is constructed, which is derived from MS-SSIM and defined as follows:

$$W^c_s = \frac{1}{N_c \cdot N'_c} \sum_{i=1}^{N_c} \sum_{j=1}^{N'_c} \text{MS-SSIM}\left(G\left(\mathbf{z}^{(i)}_c\right), \mathbf{x}^{(j)}_c\right) \quad (12)$$

where W^c_s represents the structure discrepancy between real and fake images of category c. $\mathbf{x}^{(j)}_c$ is the jth real image of the cth category, whereas $\mathbf{z}^{(i)}_c$ is the ith random noise vector whose generated image $G(\mathbf{z}^{(i)}_c)$ is distinguished as the cth category by the discriminator. N_c and N'_c are the number of real and generated images, respectively, which belongs to cth category.

The distribution-based weighting term makes the distributions of generated images as close as possible to the real ones. The structure-based weighting term forces the spatial structure of generated images to be similar to the real ones. They describe the features of generated SAR samples from two different and complementary aspects. Combining both, we obtain

$$W^c = \lambda \cdot (1 - W^c_d) + (1 - \lambda) \cdot W^c_s \quad (13)$$

where W^c is the mixed weights between real and fake images of category c, and it is also the sample weight of category c in the next epoch during training. $\lambda \in [0, 1]$ is the balanced factor to adjust the ratio between distribution and visual structure. The effect of different λ on the performance is presented in Section IV-B1.

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TABLE I

PROCESS OF CLASS ADVERSARIAL TRAINING. G AND D REPRESENT THE GENERATOR AND THE DISCRIMINATOR, RESPECTIVELY. ✓ AND × INDICATE WHETHER THE COMPONENT REACHES A DESIRED LEARNING STATE OR NOT. ↑ AND ↓ IS SIGNALS OF THE CHANGE TENDENCY (RAISE OR FALL)

	G	D	predicted class	Discrepancy	Weight	Training
state	×	×	inaccurate	large	↓	continue
	×	✓				
	✓	×				
	✓	✓	accurate	small	↑	stop

Statistically, the smaller the W_d^c is, the distribution of pixel values within generated images trends more to the real ones. Visually, the larger the W_s^c is, the appearance of the generated images are more similar to the real ones. As a combinator of both, W^c can instruct the generative modeling of the generator from the above two aspects.

In addition, as shown in the following formula (17), we punish category c with sample weight W^c to drive the generated images close to real ones. But due to the immaturity of the generator or the discriminator, several categories are absent in the generated images in the early period of training. e.g., the generator cannot generate samples in a certain category or the discriminator does not classify all samples to a certain category. In this situation, the formulas (11)–(13) are useless. To address this, a complementary sample weighting scheme is carried out, i.e., half of the smallest weight among the generated categories is assigned to the absent ones. In this way, the generator prefers to generate absent categories first and optimizes the existing generation later. A further discussion of the scheme can be seen in Section IV-C1.

B. Class Adversarial Training

Applying the sample weighting scheme to ACGAN, the class adversarial learning is carried out. It can be viewed from formulas (11)–(13) that the discrepancy between generated samples and real ones is obtained by class. If the classes of real samples are given, those of generated samples are undetermined. To settle this, the discriminator with an auxiliary classifier is used to classify the generated samples in each epoch of training. In this procedure, the predicted classes are not only determined by the discriminant of the discriminator but also influenced by the generative modeling of the generator. The class adversarial learning in DSM-ACGAN can improve both, which is described in Table I.

Table I presents the process of class adversarial training. If the generated samples are absent in a certain category or the discriminator cannot map the input to class-separable feature space, the predicted class is always inaccurate. The discrepancy between misclassified generated samples and real ones is large and its corresponding sample weight decreases. The training continues until both the generator and the discriminator are in a high-level and balanced state. As a result, the generated samples are similar to the real ones and the discriminator has the ability to assign right labels to them. Naturally, the discrepancy is small and the weight increases. The training stops when the termination criterion satisfies.

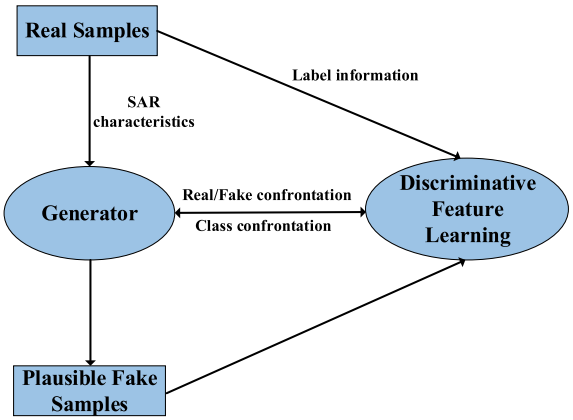


Fig. 3. Discriminative feature learning in DSM-ACGAN.

A measurable exhibition of the effectiveness of class adversarial training can be found in Table VI.

C. Discriminative Feature Learning in DSM-ACGAN

Discriminative feature learning in DSM-ACGAN strongly builds on the above class adversarial learning. Authenticity information and class information are jointly imposed on the discriminator to achieve this, which is illustrated in Fig. 3. During training, first, specific domain knowledge (SAR characteristics) from real samples promotes the generative modeling in the generator, thus obtaining plausible fake samples. Then discriminative fake samples and real samples with class information are fed to the discriminator for feature learning. The above two procedures alternate and enhance each other. It should be noted that during this processing sample weighting scheme introduce additional class adversarial training. This greatly intensifies the confrontation and facilitates discriminative feature learning in the discriminator. Finally, the well-trained discriminator with an auxiliary classifier is used to predict the class of test images.

The optimization problem for the generator can be expressed as

$$\max \quad L_C - L_S \tag{14}$$

while for the discriminator is

$$\max \quad L_C + L_S \tag{15}$$

where

$$\begin{aligned} L_S = {} & E[\log D(S = \text{real}|\mathbf{x})] \\ & + E[\log D(S = \text{fake}|G(\mathbf{z}))] \end{aligned} \tag{16}$$

$$\begin{aligned} L_C = {} & W^c \cdot E[\log A_c(C = c|D(\mathbf{x}_c))] \\ & + E[\log A_c(C = c|D(G(\mathbf{z}_c)))] \end{aligned} \tag{17}$$

where $\mathbf{x}_c$ is the real image of the cth category, $\mathbf{z}_c$ is the latent vector whose generated image $G(\mathbf{z}_c)$ is distinguished as the cth category by the discriminator. $D(\mathbf{x}_c)$ and $D(G(\mathbf{z}_c))$ are the learned discriminative features from real and generated samples, respectively. A_c represents the auxiliary classifier in the discriminator.

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Here, the balanced factor λ is a vital hyper-parameter. It controls the composition of specific domain knowledge and the tendency of discriminative feature learning. When an extreme point is selected in its range [0, 1], only one aspect of SAR characteristics is contained in the adversarial training. The two cases are named SM-ACGAN ($\lambda = 0$) and DM-ACGAN ($\lambda = 1$). If λ picks a value between 0 and 1, W_c is regarded as a weighting combination of distribution characteristics and structure characteristics. To a certain extent, the network achieves an equilibrium state between distribution modeling and structure approximation. The larger the λ is, the more the distribution considered is. On the contrary, the visual structure attracts more attention. The effect of λ can be found in Table IV, Fig. 5, Sections IV-B1 and IV-C2.

The detailed training process of DSM-ACGAN is listed in Algorithm 1, which is consistent with the mainline in Figs. 2 and 3. Among Algorithm 1, after initializing the network and sample weight vector (line 1), the training process is carried out (lines 2–18). In each epoch, the parameter of G is first fixed to update the parameter of D, the weight vector $\boldsymbol{v}$ is added to real samples according to the difference between generated samples and real ones in last epoch. Discrimination of learned features in D is constantly facilitated in this process (lines 3–8). Then the parameter of D is fixed to update the parameter of G. Meanwhile, fake samples are generated, whose amount is the same as real ones (lines 9 and 10). Finally, the generated samples are classified by the discriminator (line 11) and weight vector $\boldsymbol{v}$ is updated by class (lines 12–17). What needs to be pointed out is that k represents the number of steps applied to the discriminator. $\boldsymbol{v} \in R^{N\times 1}$ is a weight vector imposed on real samples, whose specific implementation is as formulas (13). N_c is the number of samples belonging to category c and C is the number of the category.

Algorithm 1 Adversarial Training of DSM-ACGAN

1: Initialize the generator G, the discriminator D, and the sample weight vector $\boldsymbol{v} = \{D_n\}_{n=1}^{m}$, where $D_n = 1/C$.
2: **for** number of training iterations **do**
3: **for** k steps **do**
4: Sample minibatch of m noise samples $\{z^1, z^2, \ldots, z^m\}$ from noise priori $p_z(z)$.
5: Sample minibatch of m real samples $\{\boldsymbol{x}^1, \boldsymbol{x}^2, \ldots, \boldsymbol{x}^m\}$ from training set.
6: Assign different sample weights to different categories of real samples by $\boldsymbol{v}$.
7: Continuously learn discriminative feature $D(\mathbf{x})$ and $D(G(\mathbf{z}))$ and update the parameters of the discriminator D by formula (15).
8: **end for**
9: Sample minibatch of m noise samples $\{z^1, z^2, \ldots, z^m\}$ from noise priori $p_z(z)$.
10: Generate discriminative samples $G(\mathbf{z})$ and update the parameters of the generator G by formula 14.
11: Predict the category of generated samples by the discriminator.
12: **for** $\{c \in 1, 2, \ldots, C\}$ **do**
13: Compute the distribution discrepancy W_d^c by formulas 11.
14: Compute the structure discrepancy W_s^c by formulas 12.
15: Obtain the weight W^c by formulas 13.
16: **end for**
17: Update the sample weight vector $\boldsymbol{v} = \{\{D_n^c\}_{n=1}^{N_c}\}_{c=1}^{C}$, where $D_n^c = W^c$.
18: **end for**

IV. Experimental Results and Analysis

To accurately determine vital hyper-parameters, clearly observe the effectiveness of class adversarial training and investigate the strengths of discriminative feature learning in DSM-ACGAN, several experiments are carried out. Data set introduction and experiment setting are introduced in Section IV-A. In Section IV-B, the selection of two hyper-parameters is given. Section IV-C presents the process and generated samples in class adversarial training. Experiments of three broad SAR images from different satellites are conducted in Section IV-D.

A. Data Set and Parameter Settings

Experiments are carried out on three broad SAR images acquired from F-SAR and COSMO-SkyMed. The image of F-SAR is the area of Bavaria, Germany, shown in Fig. 9(a). It has 6187 × 4278 pixels in size with an HH-polar imaging mode. The ground resolution is 1.0 m × 0.67 m. The images of COSMO-SkyMed are regions of Naples and PoDelta, Italy, shown in Figs. 10(a) and 11(a), the sizes are 16 000 × 18 332 and 16 716 × 18 308, on March 1, 2013, and September 27, 2007, respectively. They are both COSMO-SkyMed geocoded ground range level 1C products with a pixel spacing of 2.5 m, in HH-polar imaging and stripmap modes, and in X-band. All the pixels in the above three images are assigned to four terrain categories: building, water, tree, and farmland. Unknown pixels are labeled as null. The colorbar of different terrain is at the bottom of Fig. 4, and it is applied in all of the following experiments. We ignore the unknown pixels during both training and testing.

In all the following experiments, each image is divided into nonoverlapping regions for training and testing, which is able to ensure the reliability of the results. Regions from which the training and valid patches are sampled are shown in Fig. 4. For each category, we picked 6000 samples from each of the above region, among which 5000 are for training and 1000 for validation. The number of pixels and sampling ratios in each category is listed in Table II. It is clear that the sampling ratios are pretty low. The remaining regions of each scene are for testing. Limited to the size of patches, we do not set the same scales of MS-SSIM as [59] in our experiments. The number of scales in MS-SSIM is picked as three and only the first three experiential parameters in [59] are adopted in our experiments. They are normalized as follows: $\beta_1 = \gamma_1 = 0.0711$, $\beta_2 = \gamma_2 = 0.4530$, and $\alpha_3 = \beta_3 = \gamma_3 = 0.4760$. The procedure of optimization is the same as GAN, i.e., separately iteratively optimize the discriminator and the generator with backpropagation, which is shown in Algorithm 1.

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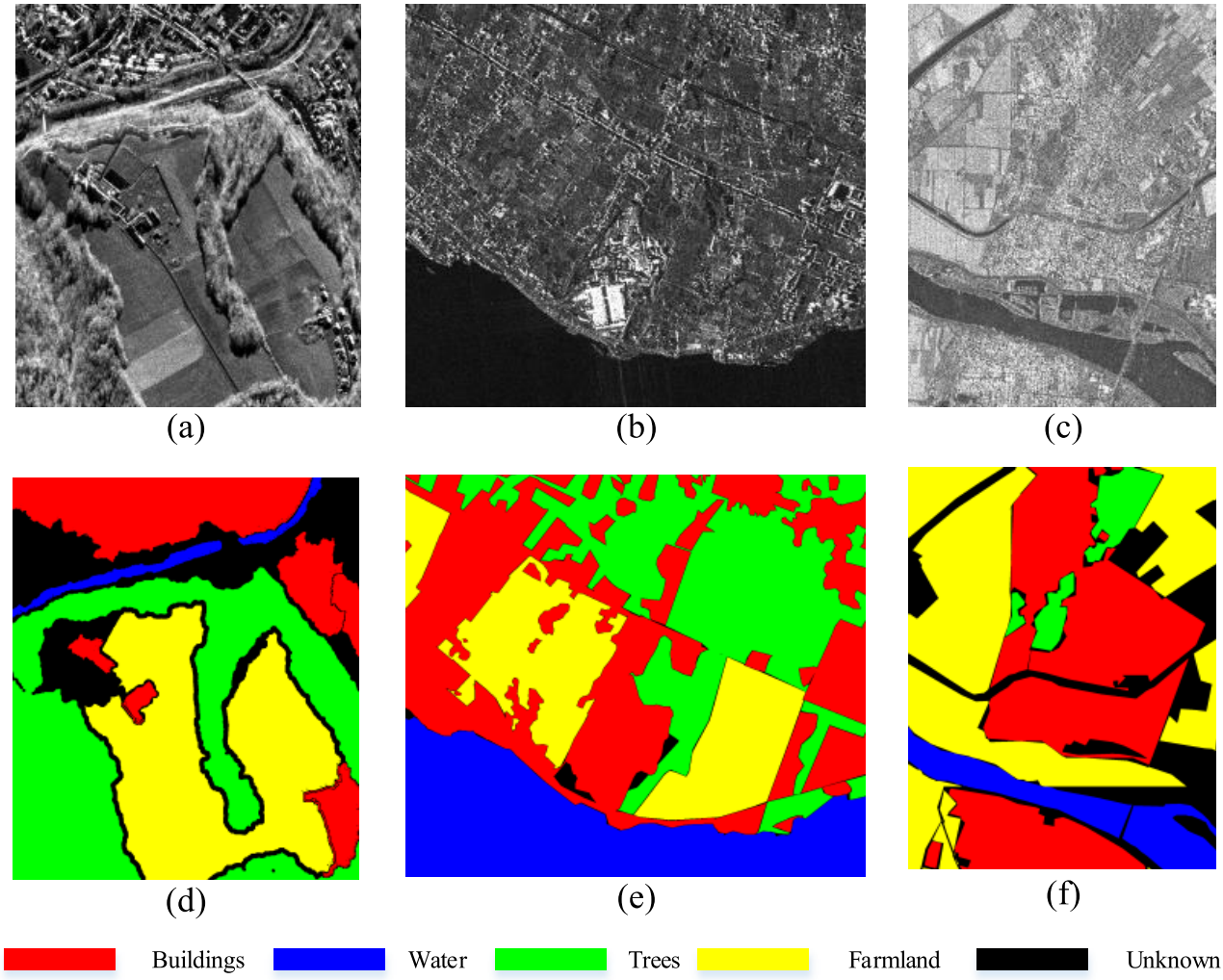


Fig. 4. Regions from which the training and validating patches are sampled. (a)–(c) Regions of Traunstein, Naples, and PoDelta, their sizes are 1200 × 1400, 1303 × 1151, and 1440 × 1902, respectively. (d)–(f) Corresponding ground truths.

TABLE II
TRAINING RATIOS FOR DIFFERENT PREDEFINED CATEGORIES OF DIFFERENT SCENES

scenes	category	pixel number	sample number	sampling ratio/‰
Traunstein	Building	2212200	5000	2.26
	Water	141567	5000	35.32
	Tree	11560967	5000	0.43
	Farmland	5600820	5000	0.89
Napoli	Building	64108325	5000	0.08
	Water	63204151	5000	0.08
	Tree	33527445	5000	0.15
	Farmland	54910258	5000	0.09
PoDelta	Building	3092052	5000	1.62
	Water	147492711	5000	0.03
	Tree	2663934	5000	1.88
	Farmland	61848977	5000	0.08

Additionally, Wasserstein distance [61] is adopted to stabilize the training. The real samples and generated samples are assigned label 1 and -1, respectively, for the source, while 1 to 4 for class. Model architecture and other training parameters can be seen in Table III.

All the experiments are implemented on the open source library Keras 2.1.3 [62] with Tensorflow 1.4.1 [63] backend. The interface is Python 2.7 and the platform is an HP Z840 workstation with an Intel Xeon(R) CPU, a GeForce GTX TITAN X GPU, and 64-GB RAM under Ubuntu 14.04 LTS.

B. Hyper-Parameter Analysis

1) Balanced Factor λ: As a key hyper-parameter in DSM-ACGAN, λ decides the composition of sample weights and training tendency of the network. Experiments are conducted to explore the relationship between λ and the performance, which is shown in Table IV and Fig. 5.

From Table IV, It can be observed that the overall accuracy (OA) of each SAR image reaches the highest value at different λ. For Traunstein, it is exactly 0.5, which indicates that a symmetrical mixture of distribution characteristics and structure characteristics brings the best guidance. On the other hand, the results of Naples and PoDelta show obvious favor. In Naples, smaller λ presents better performance, whereas larger λ contributes more in PoDelta. In summary, the results reveal that the promotion of joint statistical modeling and

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TABLE III
MODEL ARCHITECTURE AND TRAINING PARAMETERS OF DSM-ACGAN

Operation	Kernel	Strides	Feature maps	Dropout	Nonlinearity
$G_x(z)$-100×1×1 input					
Linear	N/A	N/A	1024	0.0	LeakyReLU
Transposed Convolution	5×5	2×2	256	0.0	LeakyReLU
Transposed Convolution	5×5	2×2	128	0.0	LeakyReLU
Transposed Convolution	5×5	2×2	1	0.0	Tanh
$D(x)$-1×28×28 input					
Convolution	3×3	2×2	32	0.3	LeakyReLU
Convolution	3×3	2×2	64	0.3	LeakyReLU
Convolution	3×3	2×2	128	0.3	LeakyReLU
Convolution	3×3	2×2	256	0.3	LeakyReLU
Linear[1]	N/A	N/A	1	0.0	Tanh
Linear[2]	N/A	N/A	4	0.0	Softmax
Optimizer	Adam ($lr = 10^{-4}, \beta_1 = 0.5, \beta_2 = 10^{-3}$) for $G(x)$				
	SGD ($lr = 10^{-4}, clipvalue = 0.01$) for $D(x)$				
Batch size	100				
Epoches	300				
LeakyRelu slope	0.2				
Weight, bias initialization	Isotropic gaussian ($\mu = 0, \sigma = 0.01$), Constant(0)				

* The Linear[1] and Linear[2] both connect to the last convolution layer in the discriminator $D(x)$
* Number of parameters: 8.56×10^6

TABLE IV
OA WITH DIFFERENT λ

value of λ		0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
OA(%)	Traunstein	90.03	90.11	90.16	91.47	92.56	**94.00**	93.55	92.31	91.58	90.03	89.67
	Napoli	79.53	**80.59**	79.61	79.50	79.41	79.33	79.29	79.28	78.90	78.82	78.67
	PoDelta	92.22	92.27	92.33	93.00	93.11	93.14	93.72	93.89	**94.13**	93.60	92.78

TABLE V
OA OF DIFFERENT SIZES OF SAMPLES

size of samples		12×12	16×16	20×20	24×24	28×28	32×32	36×36
OA(%)	Traunstein($\lambda = 0.5$)	84.12	86.87	90.27	91.88	**94.00**	91.20	90.71
	Napoli ($\lambda = 0.1$)	75.62	75.79	77.67	77.86	80.14	**80.59**	79.53
	PoDelta ($\lambda = 0.8$)	88.02	90.32	91.53	93.59	**94.13**	93.52	93.41

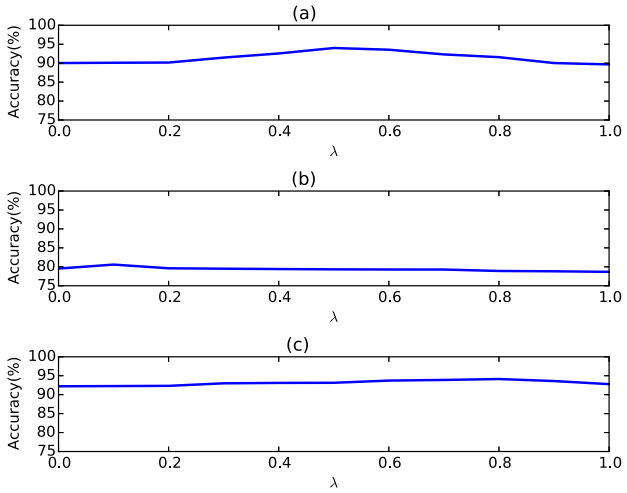


Fig. 5. Curves of different λ on OA. (a) Traunstein. (b) Napoli. (c) PoDelta.

spatial feature learning is larger than either of them. But the best value of λ is determined by the scene it applies. The curves in Fig. 5 illustrates the effect of different λ on OA. Upon observing the change tendency of different scenes, it is clear that the fluctuation of Traunstein is greater than the others. An explanation is the resolution of Traunstein is the highest, more detail information brings larger change.

2) Size of Patch: The size of patches is another common factor which affects the performance of deep models. In our experiments, their relationship is explored under the best λ's and shown in Table V. The range of patch size is from 12 × 12 to 36 × 36. It can be seen from Table V that OA increases with the size and then has a little drop. According to OA, 28 × 28 and 32 × 32 are good choices. Considering the computational complexity, 28 × 28 is picked as the standard size in all experiments.

C. Effectiveness of Class Adversarial Training

In this part, the effectiveness of class adversarial training, as well as sample weighting scheme, is presented from quantitative and qualitative aspects. They are shown in Tables VI and VII, Figs. 6 and 8, respectively. The quality of generated samples is an indicator of the effectiveness of class adversarial training and the training state of the discriminator. So we illustrate the quality of generated samples and

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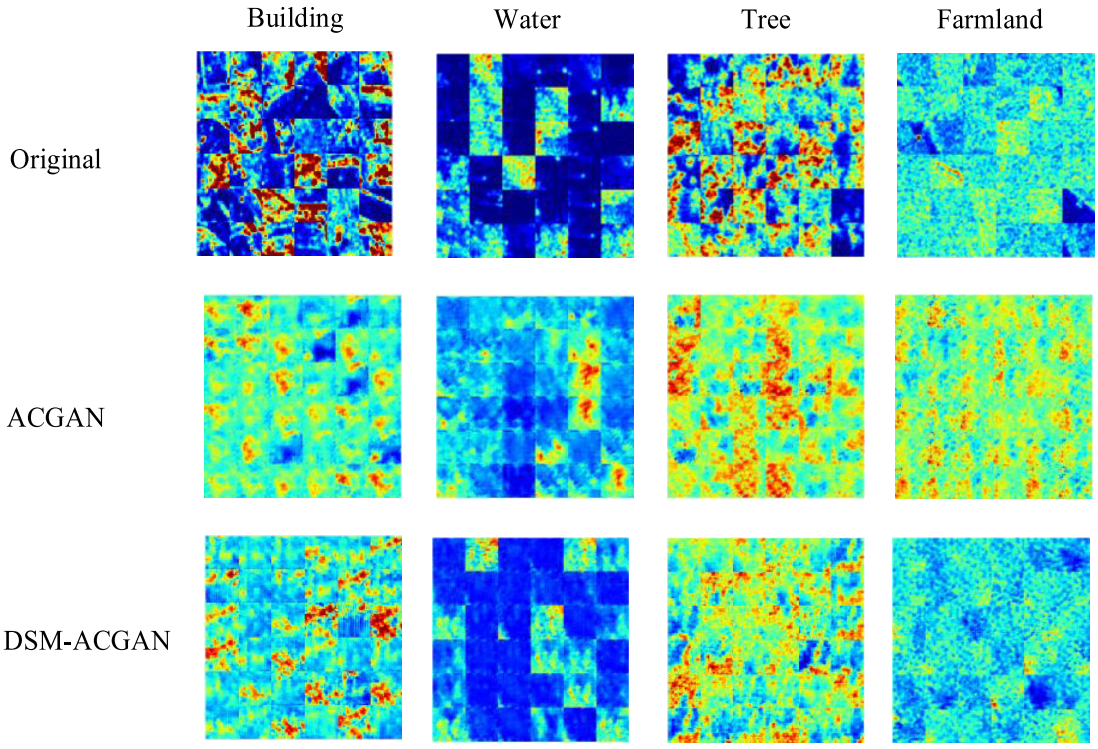


Fig. 6. Generated images in Traunstein with different methods.

TABLE VI
METRICS OF GENERATED SAMPLES IN TRAUNSTEIN

methods	category	W_{ks}^c	W_{skl}^c	W_s^c
ACGAN	Building	0.4575	8.047	0.0090
	Water	0.7881	10.918	0.0149
	Tree	0.3121	6.830	0.0065
	Farmland	0.3342	6.086	0.0013
	Mean	0.4730	7.970	0.0079
DSM-ACGAN($\lambda = 0.5$)	Building	0.3503	4.188	0.0720
	Water	0.6586	8.358	0.1317
	Tree	0.2164	6.374	0.1101
	Farmland	0.1204	5.465	0.1652
	Mean	**0.3364**	**6.096**	**0.1198**

TABLE VII
METRICS OF THE TREND OF GENERATED SAMPLES OF TREE IN TRAUNSTEIN

methods	epoch	W_{ks}^c	W_{skl}^c	W_s^c
DM-ACGAN($\lambda = 0$)	1	0.4852	6.862	0.0210
	50	0.2834	6.651	0.0652
	150	0.2484	6.503	0.0903
	300	**0.2024**	6.413	0.1002
SM-ACGAN($\lambda = 1$)	1	0.4441	7.555	0.0233
	50	0.3139	7.420	0.0876
	150	0.2923	7.335	0.0987
	300	0.2359	6.538	0.1026
DSM-ACGAN($\lambda = 0.5$)	1	0.5990	8.475	0.0217
	50	0.4823	7.412	0.0759
	150	0.2316	6.642	0.1062
	300	0.2164	**6.374**	**0.1101**

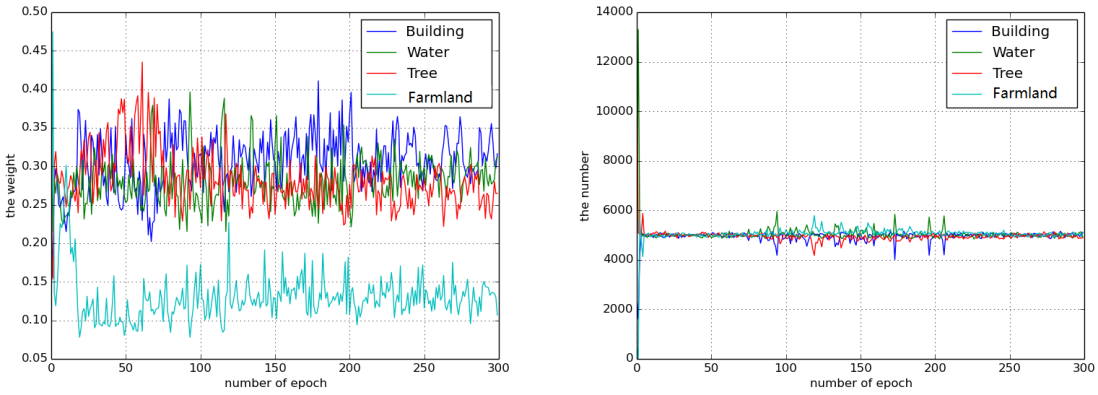
analyze the process of class adversarial training. Performance of Traunstein is taken as an example in the following parts.

1) Observation of Training Process: The change of sample weights of each terrain during training is illustrated in Fig. 7(a). As described in Section III-B, the adversarial training contains dual confrontation, which presents sharp fluctuation in curves. Investigating the generated samples of each category in Fig. 6, it is clear that farmland is the best, which corresponds to the lowest sample weight curve (cyan) at the bottom of Fig. 7(a). On the contrary, the weight curve of the building (blue) is higher than the others, confirming its harder learning process than others. This is consistent with its exhibition in Fig. 6.

Fig. 7(b) displays the number of generated images in each category during training. Two values of 0 in incipient epochs indicates the absence of class building and farmland in the generated samples. This results in large punishment on them in the next epoch. After importing the punishment, the samples of missing categories generate. The small-scale rolling in the middle is caused by the adjustment of sample weights in the adversarial learning process. In the experiments, the total number of generated samples is set the same as that of training ones, but the specific numbers are not limited. Surprisingly, all curves in Fig. 7(b) end up with approximating 5000. This is probably due to the computation of distribution discrepancy between generated samples and real ones. Particularly, formulas (1) and (2) require enough samples to precisely characterize the distribution. If any category has few generated samples, its distribution inevitably diverges from the real one, which violates the objective and will be punished. From this view, DSM-ACGAN has great potential in generating class-balanced samples.

2) Observation of Generated Samples: Fig. 6 gives an exhibition of the generated images in each category with and without sample weighting strategy. The original samples from

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(a) (b)

Fig. 7. Curves to depict the trend between two different metrics and epochs of Traunstein in DSM-ACGAN. In practice, the maximum problem of (14) and (15) is transformed to a minimum problem for convenient. So the large sample weight indicates a poor state in (a). (a) Sample weight curves in training. (b) Number of generated samples of each category in training.

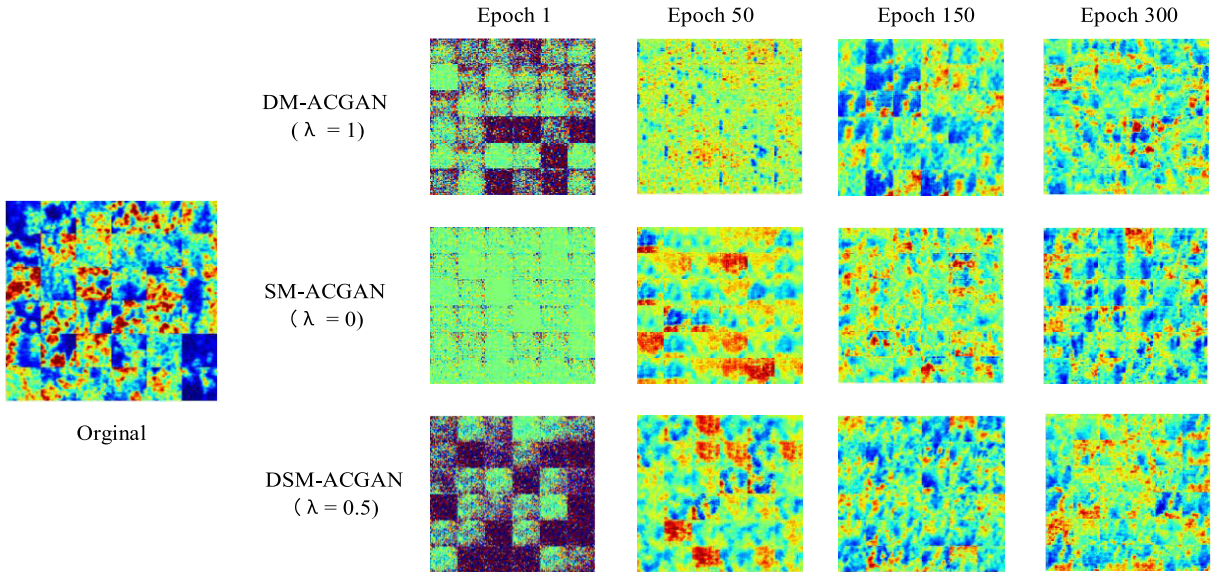


Fig. 8. Development of the generated images of tree in Traunstein.

the training set are shown in the first row, while the second and third rows are the generated samples in ACGAN and DSM-ACGAN, respectively. Each unit of one terrain displays 36 different samples in size 28 × 28. Comparing the generated images in ACGAN with those of original, it can be observed that all terrain of generated samples has an obvious gap with the original ones. As a contrast, the visual performance of generated images in DSM-ACGAN is much better than that of ACGAN. The generated samples are various and similar to the real ones, which shows the positive effect of sample weighting scheme and class adversarial training. This can also be quantitatively convinced from Table VI.

Fig. 8 and Table VII take results of category tree as an example to show the change tendency of generated samples in training. Four representative epochs 1, 50, 150, and 300 are picked to show it. Compared with tree of ACGAN shown in Fig. 6, the generated samples in epoch 300 all represent salient superiority. Further investigating the change tendency

TABLE VIII
ARCHITECTURE OF RESNET18

28×28 input
7×7, 64, stride 2
3×3, max pool, stride 2
$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}$, $\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}$
$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}$, $\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}$
$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}$, $\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}$
$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}$, $\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}$

*Number of parameters : 1.12×10^7

of these three models, it is found that DM-ACGAN trends to learning the distribution characteristic first (epoch 1 and 50 of row 2) and the scattering architecture later (epoch 150 of

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TABLE IX
CE, OE, KAPPA COEFFICIENT, AA, AND OA OF PATCH-WISE CLASSIFICATION RESULTS OF TRAUNSTEIN IMAGE WITH DIFFERENT FEATURE DESCRIPTORS

Feature descriptors	Commission error				Omission error				Kappa	AA	OA
	Building	Water	Tree	Farmland	Building	Water	Tree	Farmland			
CNN	66.47	**30.88**	46.99	11.74	63.20	96.22	12.87	33.36	0.4131	60.98	61.03
ResNet18	18.19	58.64	15.53	11.43	45.51	87.55	6.66	5.68	0.7444	74.05	85.03
PSDNN	46.58	82.97	5.41	20.74	12.87	82.99	16.52	9.76	0.7122	61.08	84.96
ACGAN	33.32	45.94	11.56	7.85	20.50	90.60	7.63	10.20	0.7689	75.33	86.79
DM-ACGAN ($\lambda = 1$)	17.80	45.23	9.67	8.12	17.98	90.55	5.02	5.69	0.8183	79.80	89.67
SM-ACGAN ($\lambda = 0$)	**6.93**	52.99	8.72	9.15	19.44	88.61	3.45	**5.31**	0.8367	80.55	90.03
DSM-ACGAN($\lambda = 0.5$)	7.77	56.66	**5.27**	**5.53**	**11.67**	**75.78**	**3.43**	5.87	**0.8923**	**81.19**	**94.00**

row 2). Although SM-ACGAN prefers to grasp the structure at the beginning (epoch 1 and 50 of row 3) and gradually draws close to its distribution (epoch 150 of row 3). But with the training continuing, this discrepancy continuously decreases. Finally, they end up with similar and desirable visual results (epoch 300 of row 2 and row 3). As for DSM-ACGAN, the learning process does not show obvious favor because the distribution and structure constraints are jointly considered. But it is worth noting that the generated images in epoch 150 have a comparable visual performance with those in epoch 300 of DM-ACGAN and SM-ACGAN, which validates the strength of DSM-ACGAN in timeliness. The tendency of metrics in Table VII is in accordance with these findings.

From another aspect, the different tendencies of generated samples with SM-ACGAN, DSM-ACGAN, and DM-ACGAN correspond to $\lambda = 0$, $\lambda = 0.5$, and $\lambda = 1$, respectively. Combining with the results in Table IV and above analysis, it can be concluded that λ has a great influence not only on the performance of discriminative feature learning, but also on the learning procedure and speed. The smaller it is, the more tendency to capture the visual appearance at first. On the contrary, the more vigor concentrates on the distribution simulation at the beginning. Among the given range, it combines both processes and accelerates the training.

D. Performance of Discriminative Feature Learning

To comprehensively verify the benefit of sample weighting strategy and class adversarial training for enhancing the discrimination of learned features, classification results of DM-ACGAN, SM-ACGAN, and DSM-ACGAN are compared with the baseline ACGAN. Besides, experiments are also compared with three state-of-the-art methods, CNN [46], ResNet18 [64], and patch-sorted deep neural network (PSDNN) [47] to show the powerful ability of DSM-ACGAN to capture high-level discriminative features. For a fair comparison, ACGAN, DM-ACGAN, SM-ACGAN and DSM-ACGAN have the same architecture and hyperparameters (except for the learning rate), which is shown in Table III. To better confirm that DSM-ACGAN truly outperforms a simple discriminator, ResNet18 is adopted and shown in Table VIII, which has a comparable amount of parameters as DSM-ACGAN. In addition, the architecture and hyperparameters of CNN are the same as that of the discriminator in DSM-ACGAN (except for the module Linear[1]) to verify the effect of adversarial training in discriminative feature learning. All the methods are trained from the beginning using SAR data and no pretrained model is adopted.

Four metrics are used to evaluate the performance of classification: OA, average accuracy (AA), kappa coefficient, commission error (CE), and omission error (OE). OA, AA, and kappa coefficient describe the global performance of classification, while CE and OE provide the detailed classification performance of each category. The definition of CE and OE are as follows.

Let $G = \{g_{ij}\}_{i,j=1}^{C}$ be the $C \times C$ confusion matrix, C is the number of the category, g_{ij} represents true class i classified to class j. For CE, ground truthing ensures that the error matrices have a higher accuracy percentage than would be the case if no pixels were ground truthed. This value is the inverse of the user's accuracy, i.e., CE $=$ 1 $-$ user's accuracy. The formula is as follows:

$$\mathrm{CE}^i = 1 - \frac{g_{ii}}{\sum_{j=1}^{C} g_{ij}}.$$

For OE, the process of ground truthing helps one to ensure that the pixel is classified correctly and the error matrices are more accurate. This value is the inverse of the producer's accuracy, i.e., OE $=$ 1 $-$ producer's accuracy. The formula is as follows:

$$\mathrm{OE}^j = 1 - \frac{g_{jj}}{\sum_{i=1}^{C} g_{ij}}.$$

1) Results of Traunstein: The data have been described in Section IV-A. Five quantitative metrics listed in Table IX give a quantitative measure of the performance. Also, the classification maps of our proposed method and several contrast methods are shown in Fig. 9.

The visual result of CNN in Fig. 9(c) is obviously inferior to other methods, which is coincident with the big accuracy gap shown in Table IX. This reveals the considerable discrepancy of learning ability between CNN and other models. Compared with ACGAN, CNN lacks the architecture of the generator, which indicates the great benefit of adversarial training to capture discriminative features. Added with class adversarial learning, DSM-ACGAN presents better performance than ACGAN. Less noisy labels in both homogeneous and heterogeneous areas can prove this. The performance of ResNet18 is obviously inferior to DSM-ACGAN, which highlights the effectiveness of our proposed method.

A great majority of the misclassification pixels in building are classified into a tree with CNN, PSDNN, and ACGAN at

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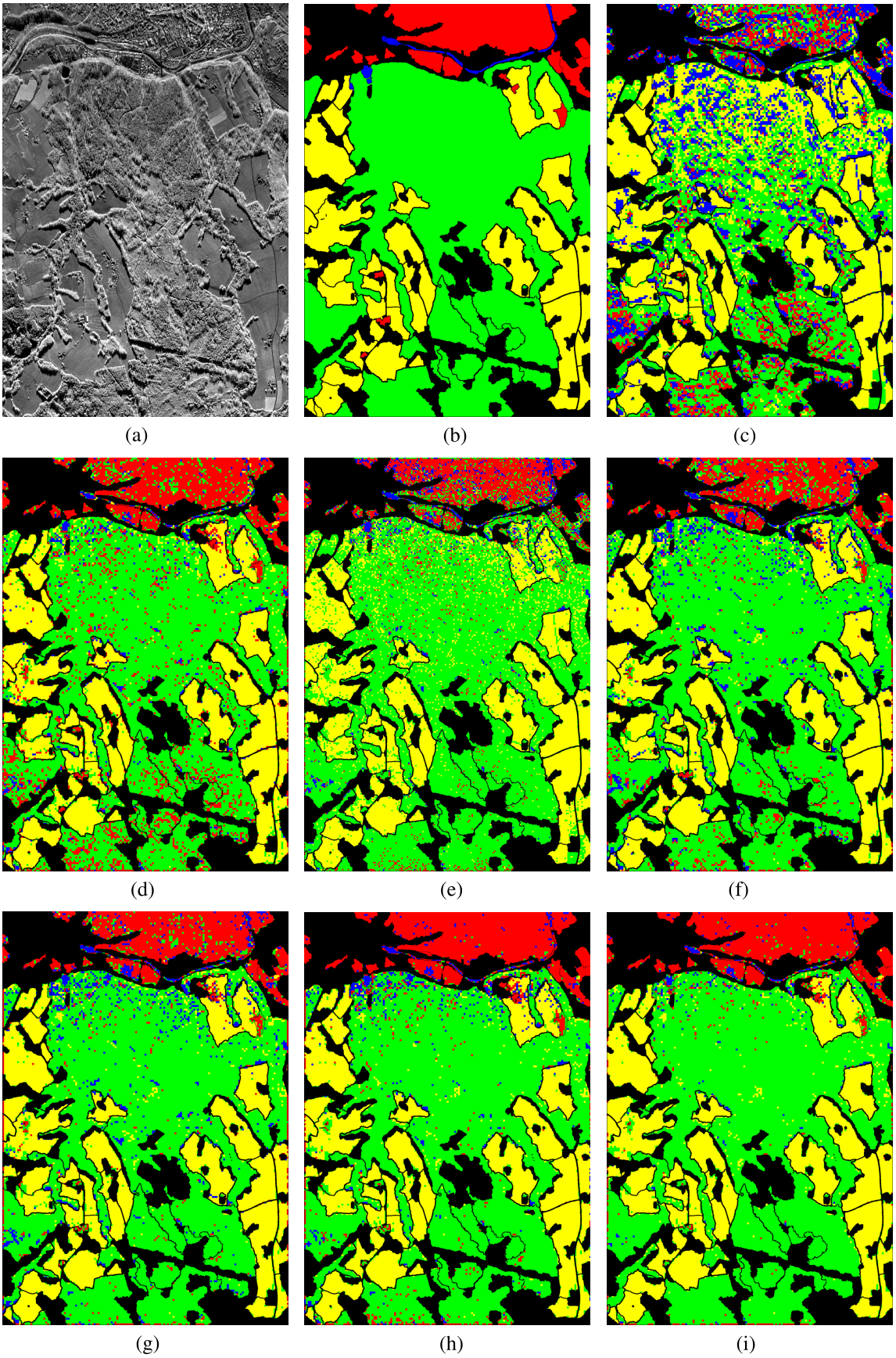


Fig. 9. Classification results of different feature descriptors in the experiments of Traunstein. (a) Original image. (b) Ground truth. (c) CNN. (d) ResNet18. (e) PSDNN. (f) ACGAN. (g) DM-ACGAN. (h) SM-ACGAN. (i) DSM-ACGAN.

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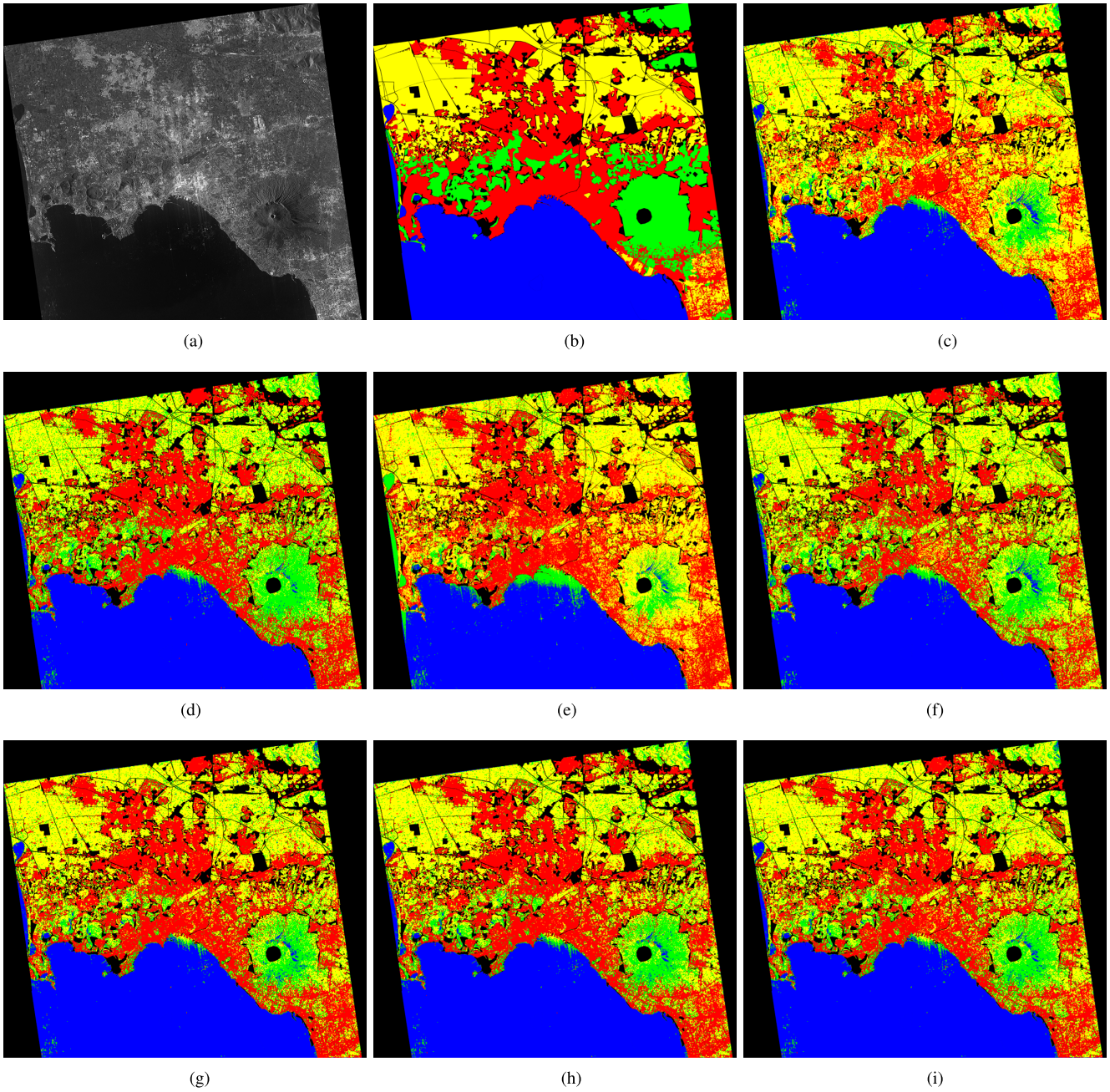


Fig. 10. Classification results of different feature descriptors in the experiments of Naples. (a) Original image. (b) Ground truth. (c) CNN. (d) ResNet18. (e) PSDNN. (f) ACGAN. (g) DM-ACGAN. (h) SM-ACGAN. (i) DSM-ACGAN.

the top of the predicted maps, whereas the results of ResNet18, DM-ACGAN, SM-ACGAN, and DSM-ACGAN categorize this area accurately. In particular, SM-ACGAN and DSM-ACGAN in Fig. 9(g) and (h) have more smooth and complete regions than others. When it comes to OA shown in Table IX, the same conclusion can be obtained. Compared with ACGAN, the OA of DSM-ACGAN are over 7% higher, especially, the CE of the building has an improvement over 25%, which is a powerful proof of the effectiveness of discriminative feature learning in DSM-ACGAN. The performance of visual results and classification are both superior to the compared ones, which highlight the advantage of the proposed framework.

2) Results of Naples: The data have been described in Section IV-A. Five quantitative metrics listed in Table X give a quantitative measure of the performance. Also, the classification maps of our proposed method and several contrast methods are shown in Fig. 10.

From Fig. 10 and Table X, it can be observed that models imposed on sample weighting scheme have better performance than state-of-the-art methods both on vision and metrics. Particularly, the OE of tree is much lower, which validates the excellence of class adversarial training. In the predicted maps, tree and farmland are confused in all classification results. Specifically, most of the tree in the top right corner is

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TABLE X

CE, OE, KAPPA COEFFICIENT, AA, AND OA OF THE PATCH-WISE CLASSIFICATION RESULTS OF NAPLES IMAGE WITH DIFFERENT FEATURE DESCRIPTORS

Feature descriptors	Commission error				Omission error				Kappa	AA	OA
	Building	Water	Tree	Farmland	Building	Water	Tree	Farmland			
CNN	36.05	2.74	73.56	**13.51**	10.00	1.38	50.49	47.54	0.6377	68.54	73.62
ResNet18	17.14	2.86	**48.25**	33.07	14.87	0.79	54.07	**31.55**	0.7046	74.67	78.16
PSDNN	19.10	8.03	81.17	17.00	15.77	**0.70**	59.92	43.25	0.6650	68.68	75.03
ACGAN	29.04	2.80	58.99	23.01	**9.29**	1.62	60.20	38.66	0.6682	71.54	75.53
DM-ACGAN($\lambda = 1$)	16.39	2.23	61.11	24.13	16.37	1.69	48.66	33.71	0.7116	74.04	78.67
SM-ACGAN ($\lambda = 0$)	15.95	2.19	57.16	23.96	15.23	1.26	47.52	32.44	0.7168	75.19	79.53
DSM-ACGAN($\lambda = 0.1$)	**12.84**	**1.95**	61.77	21.33	17.88	1.82	**43.85**	31.60	**0.7288**	**75.52**	**80.59**

TABLE XI

CE, OE, KAPPA COEFFICIENT, AA, AND OA OF THE PATCH-WISE CLASSIFICATION RESULTS OF PODELTA IMAGE WITH DIFFERENT FEATURE DESCRIPTORS

Feature descriptors	Commission error				Omission error				Kappa	AA	OA
	Building	Water	Tree	Farmland	Building	Water	Tree	Farmland			
CNN	99.91	7.48	99.99	**1.95**	99.17	0.82	99.98	20.11	0.8180	47.67	91.64
ResNet18	25.96	2.68	**16.20**	26.38	73.78	**0.73**	85.63	**1.51**	0.7899	**82.20**	90.00
PSDNN	82.37	3.10	17.60	30.88	90.26	0.84	89.44	4.57	0.7412	66.51	87.59
ACGAN	60.21	1.54	22.91	19.85	**48.45**	1.03	85.88	3.71	0.8280	73.87	92.09
DM-ACGAN($\lambda = 1$)	**25.84**	1.90	30.02	15.42	66.89	0.99	**77.75**	2.48	0.8594	81.72	92.78
SM-ACGAN($\lambda = 0$)	33.07	**1.21**	19.53	18.97	55.55	1.30	81.68	1.97	0.8462	81.81	92.22
DSM-ACGAN($\lambda = 0.8$)	35.47	1.68	26.51	13.50	58.01	1.04	77.94	2.72	**0.8718**	80.71	**94.13**

TABLE XII

TRAINING AND TESTING TIME WITH DIFFERENT METHODS ON DIFFERENT SCENES

time/mins \ scenes / methods	Traunstein(Fig.9)		Napoli(Fig.10)		PoDelta(Fig.11)	
	training	testing	training	testing	training	testing
CNN	3.71	2.01	3.32	21.15	3.56	20.98
ResNet18	5.61	2.93	5.57	23.48	5.63	24.20
PSDNN	489.95	20.11	744.83	128.81	745.16	135.31
ACGAN	238.76	2.01	240.92	21.49	236.83	21.60
DM-ACGAN	263.88	2.00	266.18	21.58	264.42	21.49
SM-ACGAN	470.77	1.98	473.85	21.86	467.95	21.39
DSM-ACGAN	485.95	1.99	489.52	21.53	484.70	21.64

misclassified as farmland. This reflects a similar backscatter between farmland and tree, which is a difficulty in discriminative feature learning.

It is worth noting that the gap between CNN, ResNet18, and DSM-ACGAN in this scene is markedly smaller than that in Traunstein. The reason may be that the resolution of Naples is lower than that of Traunstein. CNN and its variants are adept in seizing coarse features, while our models are more robust to different resolutions. To be pronounced, the AA of SM-ACGAN slightly outperforms that of DSM-ACGAN. This is caused by its lower CE of the tree, which reveals the fact that the structural feature gives more contribution than the distribution feature in this scene. It inspires a further adjustment of hyperparameter λ in classifying different terrain.

3) Results of PoDelta: The data have been described in Section IV-A. Five quantitative metrics listed in Table XI give a quantitative measure of the performance. Also, the classification maps of our proposed method and several contrast methods are shown in Fig. 11.

Compared with the visual results of CNN, ResNet18, PSDNN and ACGAN, DM-ACGAN, SM-ACGAN and DSM-ACGAN appear more consistency with the ground truth in Fig. 11(b). The main river and most discrete urban areas are accurately classified, proving the strong ability of DSM-ACGAN to capture discriminative features in complex scenes. It can also be viewed from Fig. 11(d) and (h) that the misclassification between tree and farmland of DSM-ACGAN is lower than PSDNN.

From Table XI, except for PSDNN, other methods have close OA and kappa coefficient. But with careful observation, CNN almost only divides the scene into two classes, water and farmland. The complex classes, building and tree, are missing, which indubitably confirms the much inferior learning ability of CNN. This can also be verified from its lowest value in AA.

4) Timeliness Analysis: Table XII compares the efficiency of all the methods. Training and testing time of different methods on different scenes are listed and the time is measured in minutes.

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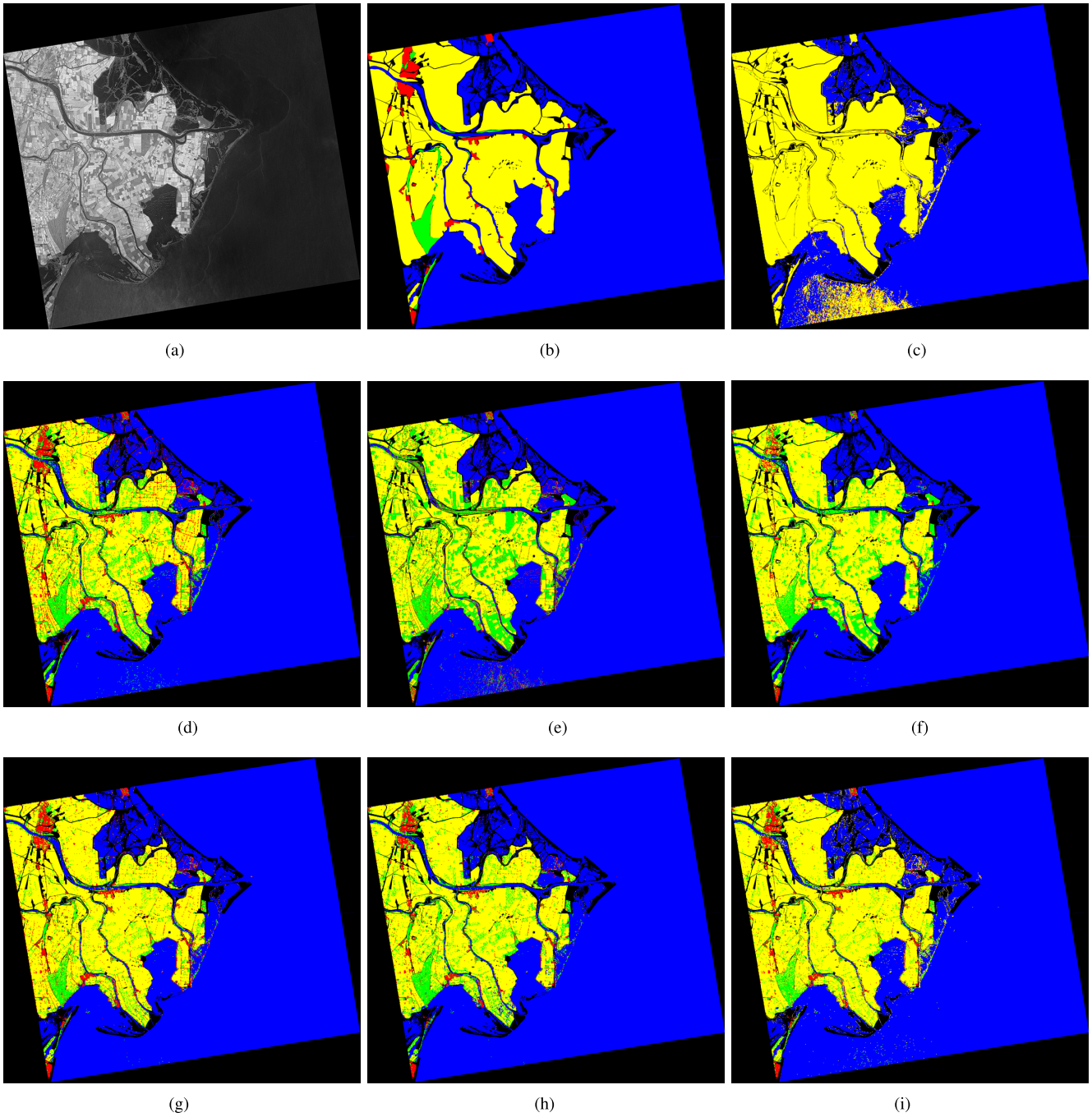


Fig. 11. Classification results of different feature descriptors in the experiments of PoDelta. (a) Original image. (b) Ground truth. (c) CNN. (d) ResNet18. (e) PSDNN. (f) ACGAN. (g) DM-ACGAN. (h) SM-ACGAN. (i) DSM-ACGAN.

From the view of training, CNN and ResNet18 have high speed due to their end-to-end architecture. The speed of PSDNN is lowest because it is layer-by-layer trained and conducted on CPU, and this is why its testing process is also inefficient. The training of GAN-based models, such as ACGAN, DM-ACGAN, SM-ACGAN, and DSM-ACGAN, is slower than CNN and ResNet18. The difficulty in optimization slows down the training, but their testing process is as fast as CNN. This is because their discriminators used for classification are the same as CNN. Comparing DM-ACGAN with SM-ACGAN, it is clear that the training of DM-ACGAN is faster than SM-ACGAN, which indicates the learning of spatial structure is harder than statistical distribution. Owing to its more parameters, the testing with ResNet18 is slightly slower than CNN.

V. Conclusion and Future Work

In this article, guided by sufficient domain knowledge of SAR images, a DSM-ACGAN is constructed to achieve comprehensive discriminative feature learning. Through class adversarial training, the generator achieves better generative modeling and the discriminator realizes effective discriminative feature learning. In the experiments, it is found that different values of balanced factor λ have a

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significant influence on the tendency and speed of class adversarial training. In addition, DSM-ACGAN shows excellent generalization in different resolutions and has a huge potential in generating class-balanced SAR images.

However, there are still some aspects can be improved in the proposed approach. As is shown in Section IV-D, different λ's achieve best results in different categories, their relationship deserves further exploration. More effective methods to capture the features of similar classes, for example, tree and farmland, is also needed.

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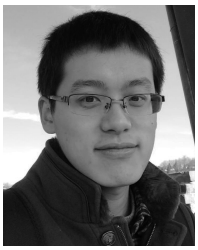
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